TEXORA · LECTURE SHEET

Advanced Machineries

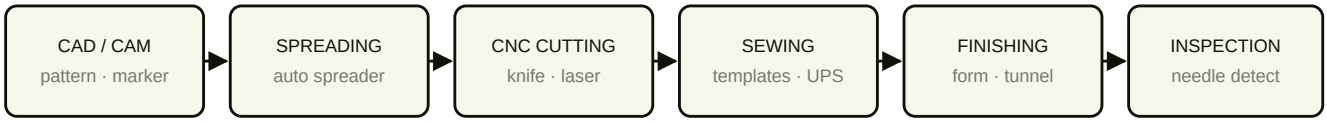
The automated machinery of a modern apparel plant — CAD and CNC cutting, template and workstation sewing, unit production systems and automatic finishing — and the output calculations that justify them.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 1 · SPECIALIZATION — APPAREL MANUFACTURING

1. What "advanced" means in a garment factory

A conventional garment factory moves material by hand at every step: patterns are drafted on paper, fabric is spread by hand and cut with a hand-guided knife, sewing operators position every ply by eye, and finished garments travel in bundles tied with string. An **advanced machinery** installation attacks the same four costs in the same order — it **computes** the pattern and marker (CAD/CAM), **mechanises** spreading and cutting (automatic spreaders, CNC cutters), **guides or performs** the sewing operation (template machines, automatic workstations), and **transports and tracks** the work automatically (unit production systems, RFID). The engineering question is never whether a machine is impressive; it is whether its cycle time, accuracy and consistency beat the manual method at the factory's order mix — which is why this course pairs every machine group with its calculation.

2. The machinery map of the plant



Material and data flow through an automated apparel plant

Figure 1 — The six machine groups of an advanced plant, in material-flow order. Every later topic of this course calculates for one of these stages.

STAGE	ADVANCED MACHINERY	WHAT IT REPLACES / IMPROVES
Pattern & marker	CAD pattern design and grading system, automatic marker-making software,	Paper drafting and manual markers; higher marker efficiency,

	plotter/printer	instant grading, digital style archive
Spreading	Automatic spreading machine with cradle feed, edge control, ply counter and tension control	Two workers pulling fabric by hand; removes tension variation and end waste, counts plies exactly
Cutting	CNC knife cutter (reciprocating blade, vacuum hold-down), laser cutter, die press	Hand-guided straight/band knife; cut accuracy to fractions of a millimetre, notches and drills placed by file
Sewing	Template sewing machines, automatic workstations (pocket setter, buttonhole, button-sew, bartack), computer-controlled lockstitch with auto trimmer and presser-foot lift	Operator eye-and-hand positioning; standardises quality, shortens training, cuts cycle time
Transport	Unit production system (UPS) / hanging garment conveyor with workstation terminals	Bundle carrying; one garment flows singly, work-in-progress and output counted automatically
Finishing	Form finisher, tunnel finisher, automatic folding and bagging machines	Hand ironing tables; consistent pressing, higher throughput per finisher
Inspection	Conveyor needle/metal detector, light-box checking stations, barcode/RFID tracking	Hand magnets and paper records; verified metal safety and real-time traceability

3. Cutting-room automation in detail

The **automatic spreader** carries the roll in a cradle or on a feed roller and lays each ply at a programmed length, aligning one edge photo-electrically and cutting the ply end square. Because the fabric is fed without pulling, stretch-sensitive fabrics are laid tension-free — the main quality gain — and the machine stops at the programmed ply count. The **CNC knife cutter** then cuts a whole lay from the marker file: a bristle bed and vacuum film hold the lay compressed while a reciprocating knife follows the pattern outlines, averaging small parts accurately enough that cut pieces no longer need re-trimming before sewing.

CUTTING METHOD	BEST USE	KEY LIMITATION

CNC reciprocating knife	Multi-ply lays of most apparel fabrics; the general-purpose choice	Blade deflection on very high lays; consumable blades and bristles
Laser	Single/few plies, intricate shapes, sealed edges on synthetics	Heat can fuse or discolour plies; slow on thick lays
Water jet	Special materials, no heat damage	Wet product, slow, high running cost — rare in apparel
Die (clicker) press	Very long runs of small identical parts (collars, cuffs, pockets)	One steel die per part/size; uneconomic for changing styles
Band knife	Accurate cutting of blocks and small parts from a pre-cut block	Operator-guided; a precision aid, not full automation

4. Sewing automation

Sewing resists full automation because fabric is limp, so the industry automates **around** the operator. A **template sewing machine** clamps the parts in a rigid template with a cut-out path; the machine sews the programmed seam while the operator loads the next template — positioning skill is replaced by template accuracy. **Automatic workstations** perform one complete operation cycle — a pocket setter folds and attaches a pocket, a buttonhole machine cuts and sews the hole and bartacks it — delivering identical results from a semi-skilled operator. The **unit production system** carries each garment on its own hanger past the workstations in sequence; the system, not a supervisor with a tally sheet, knows where every garment is and what each station produced.

WORKED EXAMPLE — TEMPLATE MACHINE VS CONVENTIONAL, SPREADING TIME

Template sewing. Attaching a shirt pocket takes 90 s by conventional lockstitch (position, sew, turn, topstitch by eye) and 45 s on a template machine (load, auto-sew, unload). In a 480-minute shift at 85% effective working, the productive time is 480 × 0.85 × 60 = 24,480 s. Output, template machine: 24,480 ÷ 45 = **544 pockets**. Conventional: 24,480 ÷ 90 = **272 pockets**. One template machine does the work of two conventional stations, with identical pocket position on every shirt.

Spreading. An automatic spreader lays a 60 m ply at 100 m/min: 60 ÷ 100 = **0.6 min per ply**. Manual spreading averages 25 m/min: 60 ÷ 25 = **2.4 min per ply**. The order of Topic 3 needs 2,000 plies: automatic 2,000 × 0.6 = 1,200 min = **20 h**; manual 2,000 × 2.4 = 4,800 min = **80 h**.

5. Reading a machine specification

Advanced machines are compared on five figures: **cycle time or speed** (stitches/min, m/min, pieces/h), **capacity** (lay height, working width, batch kg), **accuracy** (cutting tolerance, seam placement repeatability), **changeover time** (style change, template or die change) and **running cost** (power, consumables, maintenance, operator grade). A machine that wins on speed but loses an hour at every style change is the wrong machine for a high-style-mix factory — the calculation, not the brochure, decides.

Glossary

TERM	MEANING
CAD/CAM	Computer-aided design / manufacturing: digital pattern making, grading, marker making and direct drive of cutters.
CNC cutter	Computer numerically controlled cutting machine that cuts a lay directly from the marker file.
Template sewing machine	Sewing machine that follows a rigid template holding the parts, sewing a programmed seam path automatically.
Unit production system (UPS)	Overhead conveyor carrying single garments on hangers between workstations, tracking each unit electronically.
Form finisher	Pressing machine that finishes a garment on an inflatable form with steam and hot air.

Key points and summary

- Advanced machinery computes (CAD), mechanises (spreading, CNC cutting), guides (templates, workstations) and tracks (UPS, RFID) — one group per factory stage.
- Cutting choice is by material and run length: CNC knife is the general-purpose answer; die cutting wins only on long runs of identical small parts.
- Sewing automation works around the operator rather than replacing her: templates and automatic workstations standardise the operation.
- Judge machines by cycle time, capacity, accuracy, changeover and running cost together — the worked outputs (544 vs 272 pockets; 20 h vs 80 h spreading) are the deciding numbers.

Practice

Answer in your own words; brief model answers follow.

1. A bartack operation takes 24 s on an automatic bartack machine and 40 s by conventional machine with manual positioning. Using 24,480 productive seconds per shift, calculate both outputs.
2. Why is die cutting economic for collars on a long-running basic style but a poor choice for a fashion style that changes every month?
3. State the five specification figures used to compare advanced machines, and explain why changeover time matters more as order sizes shrink.
4. An automatic spreader works at 100 m/min and a manual team at 25 m/min. How long does each take to spread 500 plies of 48 m?

Answer key

1. Automatic: 24,480 ÷ 24 = 1,020 pieces. Conventional: 24,480 ÷ 40 = 612 pieces.
2. A die is a fixed cost per part and size, recovered only over a long run of identical parts. A monthly-changing style cannot amortise new dies, and the die fixes the shape while the style is still changing.
3. Speed/cycle time, capacity, accuracy, changeover time, running cost. With small orders the machine changes style more often, so hours lost at changeover eat a larger share of total production time.
4. Automatic: 500 × 48 ÷ 100 = 240 min = 4 h. Manual: 500 × 48 ÷ 25 = 960 min = 16 h.

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